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Inventor(s)	Howard; John F. et al.

Catheter with radiofrequency cutting tip and heated balloon

Abstract

Methods for draining pseudocysts and stent delivery systems for use therein are disclosed. An illustrative system may include a catheter shaft having an inflatable balloon affixed to a distal end region thereof. A cutting electrode may be disposed at the distal end of the system and at least one heating electrode may be disposed within the inflatable balloon. A self expandable stent may be disposed about the inflatable balloon. The stent may be formed of a shape memory polymer. The inflation fluid may be heated within the balloon to facilitate expansion of the stent.

Inventors: Howard; John F. (Salem, MA), Fredrickson; Gerald (Westford, MA), Regan; Kristin (Winchester, MA), Jordan; Gary A. (Litchfield, NH)

Applicant: Boston Scientific Scimed, Inc. (Maple Grove, MN)

Family ID: 53396582

Assignee: Boston Scientific Scimed, Inc. (Maple Grove, MN)

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Primary Examiner: Peffley; Michael F

Attorney, Agent or Firm: Seager, Tufte & Wickhem, LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of and claims the benefit of priority to U.S. patent application Ser. No. 15/899,985, filed on Feb. 20, 2018, now granted as U.S. Pat. No. 11,000,283 and claims the benefit of priority to, U.S. patent application Ser. No. 14/724,480, filed May 28, 2015, now granted as U.S. Pat. No. 9,913,649, which claims the benefit of priority under 35 U.S.C. § 119 to U.S. Provisional Application Ser. No. 62/004,003, filed May 28, 2014, the entire disclosures of which are herein incorporated by reference in their entirety.

TECHNICAL FIELD

(1) The present disclosure relates generally to gastric pseudocyst drainage and relates more particularly to a method for draining gastric pseudocysts and to a stent delivery system for use in said method.

BACKGROUND

(2) A gastric or pancreatic pseudocyst is a localized fluid collection, which may be rich in pancreatic enzymes, surrounded by a thin wall that may develop in the peritoneal cavity after the onset of acute pancreatitis. Although many pseudocysts resolve themselves spontaneously, some pseudocysts become quite large and require treatment due to the unwanted pressure they exert against the stomach and/or neighboring organs.

(3) One approach to treating gastric pseudocysts involves surgery and typically comprises (i) cutting through the abdominal wall of the patient to permit access to the pseudocyst through the abdominal wall, (ii) perforating or puncturing the pseudocyst, (iii) inserting a drainage tube into the pseudocyst through the perforation to allow the contents of the pseudocyst to empty through the drainage tube to a point external to the patient, (iv) removing the drainage tube from the patient once the pseudocyst has been emptied, and (v) repairing the abdominal wall. As can readily be appreciated, the surgical approach described above is invasive and has easily identifiable drawbacks associated therewith, such as an appreciable risk of infection.

(4) In some instances, an endoscopic approach to treating gastric pseudocysts may be utilized. This approach is less invasive than surgery and typically involves inserting an endoscope through the patient's mouth and into the patient's stomach. The endoscope is first used to visually locate the pseudocyst on the opposite side of the stomach wall. A needle or sphincterotome is then extended through the distal end of the endoscope to perforate both the stomach wall and the pseudocyst. A contrast agent delivered through the endoscope is then injected into the pseudocyst, and a cystogram is endoscopically performed to confirm entry into a pseudocyst (as opposed to entry into the peritoneal cavity). Following confirmation of entry into a pseudocyst, a guide wire is advanced through the endoscope and into the pseudocyst. Next, a balloon catheter is advanced through the endoscope and over the guide wire into the pseudocyst. The balloon is dilated to enlarge the perforations in the pseudocyst and the stomach and is then deflated and withdrawn. A plurality of straight endobiliary tubes or stents are then endoscopically implanted across the pseudocyst and stomach perforations to allow the contents of the pseudocyst to drain into the stomach, said biliary

tubes or stents being arranged in a side-by-side fashion and being implanted one at a time. The endoscope is then removed from the patient. When drainage is complete (typically within a few weeks), the endoscope is reintroduced into the patient, and the biliary tubes or stents are withdrawn from the patient through the endoscope using a snare.

(5) Although the aforementioned endoscopic approach has certain advantages over the surgical approach described above, the foregoing endoscopic approach still suffers from certain drawbacks. The procedural sequence may involve up to five separate instruments, exchanged through a gastroscope to complete the procedure, making the procedure complex and time consuming.

SUMMARY

(6) The disclosure is directed to several alternative designs, materials and methods of manufacturing medical device structures and assemblies.

(7) In a first example, a stent delivery system may comprise, a catheter shaft comprising an inner tubular member and an outer tubular member, the catheter shaft extending from a proximal end to a distal end, an inflatable balloon having a proximal end affixed to the outer tubular member of the catheter shaft and a distal end affixed to the inner tubular member of the catheter shaft proximal to the distal end of the catheter shaft, the inflatable balloon disposed adjacent to the distal end of the catheter shaft, a cutting electrode positioned at the distal end of the catheter shaft, a first heating electrode disposed about the inner tubular member of the catheter shaft and within an inner region of the inflatable balloon, a control console in electrical communication with each of the first cutting electrode and the second heating electrode, and a stent disposed about an outer surface of the inflatable balloon, the stent having a first collapsed configuration and a second expanded configuration.

(8) Alternatively or additionally to any of the examples above, in another example, the stent may comprise a shape memory polymer.

(9) Alternatively or additionally to any of the examples above, in another example, the cutting electrode may be disposed about the inner tubular member of the catheter shaft.

(10) Alternatively or additionally to any of the examples above, in another example, a heat shrink tube may be disposed over the cutting electrode.

(11) Alternatively or additionally to any of the examples above, in another example, the cutting electrode and the first heating electrode may be in electrical communication with the control console through separate electrical circuits.

(12) Alternatively or additionally to any of the examples above, in another example, the cutting electrode may be configured to be operated in a first cutting mode and the first heating electrode may be configured to be operated in a second heating mode.

(13) Alternatively or additionally to any of the examples above, in another example, a second heating electrode may be disposed about the inner tubular member of the catheter shaft and within the inner region of the inflatable balloon.

(14) Alternatively or additionally to any of the examples above, in another example, the expanded configuration of the stent may include a proximal retention feature and a distal retention feature.

(15) Alternatively or additionally to any of the examples above, in another example, the proximal retention feature may comprise a flared proximal end and the distal retention feature comprises a flared distal end.

(16) Alternatively or additionally to any of the examples above, in another example, the stent may be cross-linked in the first collapsed configuration.

(17) Alternatively or additionally to any of the examples above, in another example, the inflatable balloon may comprise a dumbbell shape in an inflated state.

(18) Alternatively or additionally to any of the examples above, in another example, the second expanded configuration of the stent may generally correspond to the dumbbell shape of the inflatable balloon.

(19) Alternatively or additionally to any of the examples above, in another example, the proximal

retention feature may comprise an inwardly facing flare and the distal retention feature comprises an inwardly facing flare.

(20) Alternatively or additionally to any of the examples above, in another example, the stent may be cross-linked in the second expanded configuration.

(21) Alternatively or additionally to any of the examples above, in another example, the cutting electrode may comprise an annular ring.

(22) Alternatively or additionally to any of the examples above, in another example, a stent delivery system may comprise a catheter shaft comprising an inner tubular member and an outer tubular member, the catheter shaft extending from a proximal end to a distal end, an inflatable balloon having a proximal end affixed to the outer tubular member of the catheter shaft and a distal end affixed to the inner tubular member of the catheter shaft proximal to the distal end of the catheter shaft, the inflatable balloon disposed adjacent to the distal end of the catheter shaft, a cutting electrode positioned at the distal end of the catheter shaft, a first heating electrode disposed about the inner tubular member of the catheter shaft and within an inner region of the inflatable balloon, a control console in electrical communication with each of the cutting electrode and the second electrode, a stent disposed about an outer surface of the inflatable balloon, the stent having a first collapsed configuration and a second expanded configuration.

(23) Alternatively or additionally to any of the examples above, in another example, the cutting electrode may be disposed about the inner tubular member of the catheter shaft.

(24) Alternatively or additionally to any of the examples above, in another example, a heat shrink tube may be disposed over the cutting electrode.

(25) Alternatively or additionally to any of the examples above, in another example, the cutting electrode and the first heating electrode may be in electrical communication with the control console through separate electrical circuits.

(26) Alternatively or additionally to any of the examples above, in another example, the cutting electrode may be configured to be operated in a first cutting mode and the first heating electrode may be configured to be operated in a second heating mode.

(27) Alternatively or additionally to any of the examples above, in another example, the stent may comprise a shape memory polymer.

(28) Alternatively or additionally to any of the examples above, in another example, the stent may be cross-linked in the first collapsed configuration.

(29) Alternatively or additionally to any of the examples above, in another example, the stent may be cross-linked in the second expanded configuration.

(30) Alternatively or additionally to any of the examples above, in another example, the second expanded configuration of the stent may include a flared proximal end region and a flared distal end region.

(31) Alternatively or additionally to any of the examples above, in another example, the inflatable balloon may comprise a dumbbell shape in an inflated state.

(32) Alternatively or additionally to any of the examples above, in another example, a second heating electrode may be disposed about the inner tubular member of the catheter shaft and within the inner region of the inflatable balloon.

(33) Alternatively or additionally to any of the examples above, in another example, a stent delivery system may comprise a catheter shaft comprising an inner tubular member and an outer tubular member, the catheter shaft extending from a proximal end to a distal end, an inflatable balloon having a proximal end affixed to the outer tubular member of the catheter shaft and a distal end affixed to the inner tubular member of the catheter shaft proximal to the distal end of the catheter shaft, the inflatable balloon disposed adjacent to the distal end of the catheter shaft, a first cutting electrode positioned at the distal end of the catheter shaft, a second electrode disposed about the inner tubular member and within an inner region of the inflatable balloon, a control console in electrical communication with each of the first electrode and the second electrode, and a self

expanding stent disposed about the outer surface of the inflatable balloon, the stent having a first collapsed configuration and a second expanded configuration, wherein an outer diameter of the inflatable balloon varies from the proximal end to the distal end thereof.

(34) Alternatively or additionally to any of the examples above, in another example, the inflatable balloon may have a first region having a first outer diameter, a second region having a second diameter, and a third region having a third diameter, the second diameter smaller than the first and third diameters.

(35) Alternatively or additionally to any of the examples above, in another example, the first region may be positioned adjacent and distal to the proximal end of the inflatable balloon, the third region may be positioned adjacent and proximal to the distal end of the inflatable balloon, and the second region may be positioned between the first and third regions.

(36) Alternatively or additionally to any of the examples above, in another example, the stent may comprise a shape memory polymer.

(37) Alternatively or additionally to any of the examples above, in another example, the stent may be cross-linked in the first collapsed configuration.

(38) Alternatively or additionally to any of the examples above, in another example, a method for delivering a stent from a first body lumen to a second body lumen through adjacent opposing luminal wall may comprise advancing a gastroscope having a working channel to a target location near a lumen wall of the first body lumen, advancing a stent delivery system through the working channel of the gastroscope, the stent delivery system may comprise: a catheter shaft having a proximal end and a distal end, an inflatable balloon having an inner region and an outer surface, the inflatable balloon disposed adjacent to the distal end of the catheter shaft, a first cutting electrode positioned at the distal end of the catheter shaft, a second heating electrode disposed within the inner region of the inflatable balloon, a control console in electrical communication with each of the first cutting electrode and the second heating electrode, and a shape memory polymer stent disposed about the outer surface of the inflatable balloon, the stent transformable from a collapsed state to an expanded state, contacting the lumen wall of the first body lumen with the first cutting electrode, supplying an electrical current to the first cutting electrode to create an opening in the first body lumen, contacting a lumen wall of the second body lumen with the first cutting electrode, supplying an electrical current to the first cutting electrode to create an opening in the second body lumen, disposing the inflatable balloon within the opening in the first body lumen and the opening in the second body lumen, inflating the inflatable balloon, and heating an inflation fluid within the inflatable balloon to transform the stent from the collapsed state to the expanded state within the opening in the first body lumen and the opening in the second body lumen.

(39) Alternatively or additionally to any of the examples above, in another example, inflating the inflatable balloon may comprise pre-inflating the inflatable balloon to a first pressure prior to heating the inflation fluid.

(40) Alternatively or additionally to any of the examples above, in another example, inflating the inflatable balloon further may comprise inflating the inflatable balloon to a second pressure greater than the first pressure after heating the inflation fluid.

(41) Alternatively or additionally to any of the examples above, in another example, heating the inflation fluid within the inflatable balloon may comprise supplying an electrical current to the second heating electrode.

(42) The above summary of some example embodiments is not intended to describe each disclosed embodiment or every implementation of the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The disclosure may be more completely understood in consideration of the following detailed description of various embodiments in connection with the accompanying drawings, in which:
- (2) FIG. 1 is a side view of a distal end region of an illustrative stent delivery system.
- (3) FIG. 2 is a cross-sectional view of the illustrative stent delivery system of FIG. 1.
- (4) FIGS. 3A-3F are fragmentary schematic views illustrating how the stent delivery system of FIG. 1 may be used.
- (5) FIG. 4 is a side view of a distal end region of another illustrative stent delivery system.
- (6) FIG. 5 is a perspective view of another illustrative stent.
- (7) FIG. 6 is a perspective view of another illustrative stent.
- (8) FIG. 7 is a perspective view of another illustrative stent.
- (9) FIG. 8 is a perspective view of another illustrative stent.
- (10) FIG. 9 is a perspective view of another illustrative stent.
- (11) FIG. 10 is a perspective view of another illustrative stent.
- (12) While the disclosure is amenable to various modifications and alternative forms, specifics thereof have been shown by way of example in the drawings and will be described in detail. It should be understood, however, that the intention is not to limit aspects of the disclosure to the particular embodiments described. On the contrary, the intention is to cover all modifications, equivalents, and alternatives falling within the scope of the disclosure.

DETAILED DESCRIPTION

- (13) For the following defined terms, these definitions shall be applied, unless a different definition is given in the claims or elsewhere in this specification.
- (14) All numeric values are herein assumed to be modified by the term “about”, whether or not explicitly indicated. The term “about” generally refers to a range of numbers that one of skill in the art would consider equivalent to the recited value (i.e., having the same function or result). In many instances, the term “about” may be indicative as including numbers that are rounded to the nearest significant figure.
- (15) The recitation of numerical ranges by endpoints includes all numbers within that range (e.g., 1 to 5 includes 1, 1.5, 2, 2.75, 3, 3.80, 4, and 5).
- (16) Although some suitable dimensions ranges and/or values pertaining to various components, features and/or specifications are disclosed, one of skill in the art, incited by the present disclosure, would understand desired dimensions, ranges and/or values may deviate from those expressly disclosed.
- (17) As used in this specification and the appended claims, the singular forms “a”, “an”, and “the” include plural referents unless the content clearly dictates otherwise. As used in this specification and the appended claims, the term “or” is generally employed in its sense including “and/or” unless the content clearly dictates otherwise.
- (18) The following detailed description should be read with reference to the drawings in which similar elements in different drawings are numbered the same. The detailed description and the drawings, which are not necessarily to scale, depict illustrative embodiments and are not intended to limit the scope of the invention. The illustrative embodiments depicted are intended only as exemplary. Selected features of any illustrative embodiment may be incorporated into an additional embodiment unless clearly stated to the contrary.
- (19) A gastric or pancreatic pseudocyst is an aggregation of tissue, fluid, debris, pancreatic enzymes and blood that often develops in the peritoneal cavity after the onset of acute pancreatitis. Although many pseudocysts resolve themselves spontaneously, some pseudocysts become quite large and require treatment due to the unwanted pressure they exert against the stomach and/or neighboring organs. In some instances, it may be desirable to create an opening in the pseudocyst and the adjacent gastrointestinal tract and place a drainage stent therebetween to allow the

pseudocyst to drain into the gastrointestinal tract. A single catheter device may be used to create an incision through a gastric structure (such as the stomach or duodenum) and the pseudocyst, dilate the incision, and place a large diameter stent within the incision. While the devices and methods described herein are discussed relative to pseudocyst drainage, it is contemplated that the devices and methods may be used in other locations and/or applications where placement of a stent is desired.

(20) FIG. 1 is a side view of a distal end region of an illustrative stent delivery system **10**. The stent delivery system **10** may include an elongate catheter shaft **12** having a proximal end (not shown) and a distal end **14**. The catheter shaft **12** may extend proximally from the distal end **14** to the proximal end which is configured to remain outside of a patient's body. Although not shown, the proximal end of the catheter shaft **12** may include a hub attached thereto for connecting other treatment devices or providing a port for facilitating other treatments. The distal end **14** may include an introducer tip **22**. It is contemplated that the stiffness and size of the catheter shaft **12** may be modified to form a delivery system **10** for use in various locations within the body. The catheter shaft **12** may further define a lumen **16** (shown in FIG. 2) through which a guidewire (not explicitly shown) may be passed in order to advance the catheter to a predetermined position, although this is not required. The stent delivery system **10** may be configured to be advanced through a working channel of an endoscope, gastroscope or other guide means **18**.

(21) The stent delivery system **10** may further include an inflatable balloon **20** affixed adjacent to a distal end region **26** of the catheter shaft **12**. An expandable stent **24** may be disposed over the balloon **20**. As will be discussed in more detail below, the shape of the expanded balloon **20** may be customized to achieve a desired shape of the expanded stent **24**, although this is not required. In some instances, the outer diameter of the balloon **20** in an inflated configuration may vary over the length of the balloon **20**. For example, the balloon **20** may include a first region **28** having a first outer diameter, a second region **30** having a second outer diameter, and a third region **32** having a third outer diameter. In some instances, the first and third regions **28**, **32** may have larger diameters than the second intermediate region **30**. This may create a balloon **20** having generally enlarged regions adjacent the proximal end region **34** and the distal end region **36** with a reduced diameter portion in an intermediate region **30** therebetween or a having a generally dumbbell shape.

(22) FIG. 2 illustrates a cross-sectional view of the distal end region of the illustrative stent delivery system **10** of FIG. 1. The catheter shaft **12** may include an outer tubular member **38** and an inner tubular member **40**. A proximal waist **42** of the balloon **20** may be secured to a distal end region **56** of the outer tubular member **38**. A distal waist **44** of the balloon **20** may be secured to a distal end region **58** of the inner tubular member **40**. The inner tubular member **40** may extend distally beyond the distal waist **44** of the balloon **20**. In some instances, an annular inflation lumen **60** may be disposed between the outer tubular member **38** and the inner tubular member **40**. The inflation lumen **60** may allow inflation fluid to pass from an inflation fluid source configured to remain outside the body to the interior region **62** of the balloon **20**.

(23) The stent delivery system **10** may further include one or more heating electrodes **46a**, **46b** (collectively referred to hereinafter as **46**) disposed about the inner tubular member **40**. While the system **10** is illustrated as including two heating electrodes **46**, it is contemplated that the system **10** may include any number of heating electrodes **46** desired, such as, but not limited to one, two, three, four, or more. The electrodes **46** may be configured to receive radiofrequency (RF) energy from an RF generation console or control console **15** configured to remain outside the body to heat the inflation fluid within the interior region **62** of the balloon **20** using an applied voltage such as, but not limited to a continuous, discontinuous, intermittent, or pulsed voltage. In some instances, an algorithm may be used to control the voltage applied. In some instances, the voltage applied to the electrodes **46** may be different (e.g. less than or greater than) from the voltage applied to the cutting electrode **48**. In some embodiments, a continuous lower voltage algorithm may be used to facilitate deployment of the stent **24**. The electrodes **46** may be connected to the RF generation console

through one or more electrical conductors **17**, such as, but not limited to the copper lead of a copper-constantan thermocouple (T/C). In some instances, the electrodes **46a**, **46b** may be connected to the RF generation console such that they may be individually controlled. In other instances, the electrodes **46** may be simultaneously controlled. It is further contemplated that the inflation fluid may be heated by other means including application of thermal, ultrasound, laser, microwave, and other related energy sources and these devices may require that power be supplied by the generator in a different form.

(24) The balloon **20** may further include an expandable stent **24** disposed about an outer surface of the balloon **20**. The stent **24** may be expandable from a first collapsed configuration (not shown) to a second expanded configuration. The stent **24** may be structured to extend between a pancreatic pseudocyst and the gastrointestinal tract to allow for drainage of the pseudocyst. In some instances, in the expanded configuration, the stent **24** may include retention features or anti-migration flares **50**, **52** positioned adjacent to the proximal end **66** and the distal end **68** of the stent **24**. The anti-migration flares **50**, **52** may be configured to engage an interior portion the walls of the pseudocyst or the gastric structure. In some embodiments, the retention features **50**, **52** may have a larger diameter than an intermediate portion **54** of the stent **24** to prevent the stent **24** from migrating once placed between the pseudocyst and the gastrointestinal tract. It is contemplated that the retention features **50**, **52** may have a larger cross-sectional area than the incision in the pseudocyst wall or the wall of the gastric structure.

(25) In some embodiments, the stent **24** may be a self expanding stent formed from a shape memory polymer (SMP). However, it is contemplated that the stent **24** may be formed from any material desired. In broad terms, shape memory polymers behave similarly to shape memory alloys such as the nickel-titanium alloys commonly referred to as Nitinol. Shape memory polymers may be formed in a parent (or remembered) shape. The shape memory polymer may be temporarily deformed into another shape by heating the polymer above the transition temperature (in some instances this may be the glass transition temperature or the melting temperature), changing the shape of the polymer, and cooling the polymer while maintaining it in the temporary shape. An external stimulus, such as, but not limited to, heat, may be used to return the shape memory polymer to the remembered shape from the temporary shape. The shape memory polymer may be selected to be biocompatible and/or bioabsorbable. A bioabsorbable stent may be designed to completely drain the pseudocyst and then deteriorate soon after which may eliminate the need for a removal procedure. In some instances, the stent **24** may be formed from polycyclooctene. In other instances, a blend of polycyclooctene and poly(styrene-butadiene-styrene) (SBS) may be utilized. These are just examples. It is contemplated that the stent **24** may be formed or cross-linked in either the desired expanded configuration or in a generally collapsed configuration. The stent **24** may be formed to have generally solid walls. This may prevent tissue in-growth as well as facilitate removal of the stent **24**. In some instances, a SMP stent may also offer high crush resistance allowing a single stent to be used. It is contemplated that the stent **24** may be compressed and/or elongated into a small enough diameter to fit through the lumen of the gastroscope **18** while also being capable of expanding to a large enough diameter, up to 18 French (6 millimeters) or more, such that a single stent **24** may provide the desired drainage.

(26) In some embodiments, the stent **24** may be formed or cross-linked in its collapsed or unexpanded shape making this the remembered shape. The unexpanded stent **24** may then be loaded over the uninflated balloon **20** (not explicitly shown), which may then be brought through the working channel of a gastroscope **18**. Once the loaded balloon **20** is clear of the gastroscope **18**, it may be positioned in the correct location relative to the pseudocyst. The balloon **20** may then be filled with saline or other inflation fluid and heated using heating electrodes **46**. When heated above a critical transition temperature value, the SMP material becomes malleable. As the balloon **20** is inflated under the heated SMP material, the stent **24** may expand with the balloon **20**, thus taking on the shape of the balloon **20**. For example, the enlarged first and third regions **28**, **32** of the

balloon **20** may form anti-migration flares **50**, **52** while the intermediate region **30** of the balloon **20** may form a large diameter lumen configured to be positioned between the pseudocyst and the gastric structure. While the balloon **20** remains inflated, the heating source may be turned off, allowing the balloon **20**, stent **24** and inflation fluid to cool. Once the stent **24** has cooled, it may become stiff and retain its temporary or deformed shape. The balloon **20** may then be deflated and removed, leaving the expanded stent **24** in the correct drainage location, as will be discussed in more detail with respect to FIGS. **3A-3F**. It is contemplated that stents formed in the unexpanded configuration may be reshaped using different size balloons allowing the shape of the expanded stent to be customized to a desired application. It is contemplated that the stent **24** may be removed using the reverse of the placement process. For example, the deflated balloon **20** may be brought through the gastroscope **18** and then placed in the center of the expanded stent **24**. The balloon **20** may be inflated with an inflation fluid, so that the outer surface of the balloon **20** contacts an inner surface of the stent **24**. The balloon may be heated using, for example, heating electrodes **46** and resultantly heating the stent **24**. The stent **24** may return to its malleable state. The balloon **20** may then be deflated. Since the remembered shape of the stent **24** is the unexpanded size, it reverts back to its original state as the balloon **20** is deflated. The stent **24** once again fits over the balloon **20**, and can be removed by pulling the whole device through the gastroscope **18**.

(27) In other embodiments, the stent **24** may alternatively be formed by injection molding SMP material in the expanded configuration, with a large diameter and anti-migration features. The stent **24** may be cross-linked in this state, meaning that the expanded configuration is the remembered shape of the stent **24**. The stent **24** may be heated and compressed over an uninflated balloon **20** (not explicitly shown), such that it fits through the working channel of a gastroscope **18**. In some instances, the stent **24** may be stretched along its longitudinal axis and radially compressed about the balloon **20** such that the collapsed delivery configuration has a longer length and smaller cross-section than the expanded configuration. Stretching the stent **24** along its longitudinal axis may reduce the diameter of the stent **24** more than would occur if the stent **24** were only radially compressed. It is contemplated that this may allow more material to be used in forming the stent **24**. Longitudinal stretching and/or radial compression may allow the anti-migration features **50**, **52** to flatten and for the stent **24** to become sufficiently small to pass through the gastroscope **18**. Once the balloon **20** passes completely through the gastroscope **18**, and is placed in the correct deployment location, the balloon **20** may be inflated with an inflation fluid and heated above the transition temperature of the SMP, using, for example, heating electrodes **46**. The SMP material may become malleable when heated and revert to its predetermined shape (which it was set to by cross linking before the stent **24** was compressed onto the balloon **20**). The stent **24** is then cooled by allowing the balloon **20** to cool. The balloon **20** is then deflated and the device **10** is removed through the gastroscope **18**, leaving the stent **24** expanded and in the correct position. It is contemplated that when the stent **24** is cross-linked in the expanded configuration, the balloon **20** used to deploy the stent **24** may be straight or have a relatively constant diameter over the first region **28**, the second region **30**, and the third region **32**. It is further contemplated that the stent **24** may be removed using the reverse of the placement process. The deflated balloon **20** may be brought through the gastroscope **18** and then placed in the center of the expanded stent **24**. The balloon **20** may be inflated with an inflation fluid, so that the outer surface of the balloon **20** contacts an inner surface of the stent **24**. The balloon **20** may be heated using, for example, heating electrodes **46** and resultantly heating the stent **24**. The stent **24** may return to its malleable state. A retrieval sheath (not shown) or other retrieval mechanism may be utilized to compress the stent **24** into a size and shape suitable for removal. The balloon **20** may then be deflated and allowed to cool. The stent **24** may once again fit over the balloon **20**, and can be removed by pulling the whole device through the gastroscope **18**.

(28) A cutting electrode **48** may be disposed about the inner tubular member **40** adjacent the distal end **14** of the catheter shaft **12**. The cutting electrode **48** may be an annular ring having an inner

diameter similar to an outer diameter of the inner tubular member **40**. A distal end **64** of the cutting electrode **48** may generally align with the distal end **14** of the catheter shaft **12** such that only the distal edge of the cutting electrode **48** is exposed. The introducer tip **22** may be heat shrunk over the cutting electrode **48** and the distal end region **58** of the inner tubular member **40** to insulate the cutting electrode **48** as well as to dilate an opening formed by the cutting electrode **48**. It is contemplated that the introducer tip **22** may be tapered to allow the distal edge of the cutting electrode **48** to be exposed while also insulating the remaining portion of the electrode **48**. In some embodiments, the cutting electrode **48** may be a 316 stainless steel ring. It is contemplated that, in some instances, the electrode **48** may be a radiopaque marker ring or may include radiopaque properties. The electrode **48** may be configured to receive radiofrequency (RF) energy from an RF generation console or control console **15** to create an incision in a body tissue using a pulsed high voltage RF algorithm. The cutting electrode **48** may be operated in a monopolar mode with a grounding electrode provided on the skin of the patient to provide a return grounding path for the RF energy. The electrode **48** may be connected to the console through one or more electrical conductors **19**, such as, but not limited to the copper lead of a copper-constantan thermocouple (T/C). It is contemplated that the cutting electrode **48** may be in communication with the same RF generation console as the heating electrodes **46**, although this is not required. When so provided, a single RF generation system may be configured to effect the cutting operation independently of the heating operation with the addition of a second set of control parameters and separate electrical circuits. It is contemplated that the RF generation console may be custom built for this application, or could be a standard clinical system. It is further contemplated that the incision may be created by other means including application of thermal, ultrasound, laser, microwave, and other related energy sources and these devices may require that power be supplied by the generator in a different form.

(29) The cutting electrode **48** may allow the stent delivery system **10** to create an incision through the gastric structure and the pseudocyst without withdrawal or exchange of devices which may save time and reduce the number of devices required for this procedure. It is further contemplated that a mechanical blade or needle distal to the balloon may require the device to be protected during tracking through the scope and positioning, and mechanically extended beyond its protection to make the incision thus requiring additional devices and/or withdrawal of devices as the presence of the balloon prevents incorporation of mechanical protection and actuation means to effect this.

(30) Referring now to FIGS. **3A-3F** there is illustrated an illustrative method in which the system **10** may be used to drain a gastric pseudocyst. In FIG. **3A**, a gastric pseudocyst **100** is shown pressing inwardly against the exterior surface of a wall **102** of the stomach **104**. However, it is contemplated that the pseudocyst may be pressed up against other portions of the gastrointestinal tract. The patient may be prepped and a standard patient return ground pad applied adhesively to the patient. The gastroscope **18** may be advanced through the mouth of the patient and brought to a predetermined location adjacent a lumen wall within a first body lumen near the pseudocyst, as shown in FIG. **3B**. The gastroscope **18** may be precisely located by visual observation of the distention caused in the stomach or duodenal wall by the pseudocyst and/or fluoroscopy. The stent delivery system **10** may be introduced through the working channel of the gastroscope. The distal tip **14** including the distal end **64** of the cutting electrode **48** may be positioned against the lumen wall or stomach wall **102**. RF energy is applied to the cutting electrode **48** from the RF generator in a first cut mode using an applied voltage such as, but not limited to a continuous, discontinuous, intermittent, or pulsed voltage. In some instances, an algorithm may be used to control the voltage applied. In some instances, the voltage applied to the cutting electrode **48** be different (e.g. less than or greater than) from the voltage applied to the heating electrodes **46**. For example, a pulsed high voltage algorithm may be used. The distal end **64** of the cutting electrode **48** may cut through the stomach wall **102** with light pressure applied on the catheter and subsequently come into contact with a second lumen wall or the pseudocyst **100** wall. With continued application of RF

energy and light pressure applied on the catheter **12**, the distal end **64** of the cutting electrode may cut through the second lumen wall or pseudocyst **100** to create an opening or incision **106** (shown in FIG. 3C) spanning from the stomach **104** to the pseudocyst **100**. This may result in release of cyst contents into the gastrointestinal tract, where they may be aspirated. Patients qualifying for endoscopic drainage may have adherences between the pancreas and the adjacent gastric structure (stomach or duodenum) which prevents leakage of the pseudocyst material into the peritoneal cavity. Continued pressure on the tip begins the dilation process as the insulating tip **22** enters the incision caused by the RF cut.

(31) The catheter **12** may be advanced until the stent **24** is properly located in the pseudocyst **100** as illustrated in FIG. 3C. The stent **24** can incorporate radiopaque characteristics to facilitate verification of placement. The RF generator mode may be switched to a second heat mode for deployment of the stent **24**. The balloon **20** may be slightly pre-inflated with inflation fluid to a first pressure (gauge pressure) of, for example, 2-4 atmospheres (atm) (202-405 kilopascals, kPa) and the heat mode actuated thus heating the inflation fluid in the balloon **20** by passing RF energy between the two heating electrodes **46a**, **46b** in a bipolar mode. The balloon **20** may be further inflated to a second pressure (gauge pressure) in the range of, for example, 10 atm (1013 kPa), adding inflation fluid as the indicated temperature approaches the desired target transition temperature. In some instances, a separate temperature sensor or thermocouple may be provided within the balloon **20** while in other instances the electrodes **46** may be used to measure the temperature of the inflation fluid. The stent **24** may expand in response to the heat and pressure until the balloon **20** is fully inflated, which may be indicated by stable pressure. Inflation of the balloon **20** may simultaneously dilate the incision **106** and expand the stent **24**, as shown in FIG. 3D.

(32) As discussed above, the stent **24** can be formed in either the expanded configuration or the collapsed configuration. When the stent **24** is formed in the expanded configuration, application of heat to the transition temperature will cause the stent **24** to regain its “remembered” expanded state. When the stent **24** is formed in the collapsed configuration, application of heat and pressure from the expanding balloon **20** will cause the stent **24** to be deformed into a “temporary” shape following the profile of the inflated balloon **20**. The RF energy may be turned off and the system is allowed to cool to body temperature effectively “freezing” the stent **24** in its expanded state. The balloon **20** may be deflated while the stent **24** remains in the expanded configuration, as shown in FIG. 3E, to establish a drainage path from the pseudocyst **100** to the stomach **104** or gastrointestinal tract. As can be seen, a distal anti-migration flare **52** may be positioned within the pseudocyst and a proximal anti-migration flare **50** may be positioned with the stomach **104**. The anti-migration flares **50**, **52** may have a cross-sectional area larger than a cross-sectional area of the incision **106**. This may prevent or help prevent the stent **24** from migrating into either the pseudocyst **100** or the stomach **104** once the stent **24** has been deployed. The catheter **12** may then be withdrawn through the gastroscope **18** as shown in FIG. 3F. It is contemplated that in some instances, the RF generator may include a manual control sequence. In other instances, the RF generator may include some automatic control sequences. For example, while each energy application (cut or heat mode) may be initiated manually, the energy termination may be manual or automatic.

(33) FIG. 4 is a side view of a distal end region of another illustrative stent delivery system **200**. The stent delivery system **200** may be similar in form and function to the stent delivery system **10** described above. The stent delivery system **200** may include an elongate catheter shaft **202** having a proximal end (not shown) and a distal end **204**. The catheter shaft **202** may extend proximally from the distal end **204** to the proximal end which is configured to remain outside of a patient's body. Although not shown, the proximal end of the catheter shaft **202** may include a hub attached thereto for connecting other treatment devices or providing a port for facilitating other treatments. The distal end **204** may include an introducer tip **210**. It is contemplated that the stiffness and size

of the catheter shaft **202** may be modified to form a delivery system **200** for use in various locations within the body. The catheter shaft **202** may further define a lumen through which a guide wire (not explicitly shown) may be passed in order to advance the catheter to a predetermined position, although this is not required. The stent delivery system **200** may be configured to be advanced through a working channel of an endoscope, gastroscope or other guide means **208**.

(34) The stent delivery system **200** may further include an inflatable balloon **212** affixed adjacent to a distal end region **206** of the catheter shaft **202**. The catheter shaft **202** may include an outer tubular member and an inner tubular member. A proximal waist **218** of the balloon **212** may be secured to a distal end region of the outer tubular member. A distal waist of the balloon **212** may be secured to a distal end region of the inner tubular member. The inner tubular member may extend distally beyond the distal waist of the balloon **212**. In some instances, an annular inflation lumen may be disposed between the outer tubular member and the inner tubular member. The inflation lumen may allow inflation fluid to pass from an inflation fluid source configured to remain outside the body to the interior region of the balloon. While not explicitly shown, the stent delivery system **200** may further include one or more heating electrodes disposed within the balloon **212** and a cutting electrode disposed adjacent the distal end **204**.

(35) In some instances, the diameter of the balloon **212**, in an inflated configuration, may vary over the length of the balloon **212**. For example, the balloon **212** may include a proximal flared region **220a** and a distal flared region **220b**. The flared regions **220a**, **220b** may have a larger cross-sectional area than an intermediate region **224** disposed between the proximal flare **220a** and the distal flare **220b**. This may create a balloon **212** having generally enlarged regions adjacent the proximal end region **214** and the distal end region **216** with a reduced diameter portion in an intermediate region **224** therebetween or a having a generally dumbbell shape.

(36) It is contemplated that the transition **222a**, **222b** from the cross-sectional area of the intermediate region **224** to the flared regions **220a**, **220b** may be gradual, sloped, or occur in an abrupt step-wise manner, as desired. The shape of the balloon **212** in the inflated state may be selected to mold a shape memory stent into a desired configuration. While not explicitly shown, an unexpanded shape memory stent may be loaded over the uninflated balloon **212**. The stent delivery system **200** may then be brought through the working channel of a gastroscope **208**. Once the loaded balloon **212** is clear of the gastroscope **208**, it may be positioned in the correct location relative to the pseudocyst. The balloon **212** may be filled with saline or other inflation fluid and heated using heating electrodes (not explicitly shown). When heated above a critical transition temperature value, the SMP material becomes malleable. As the balloon **212** is inflated under the heated SMP material, the stent may expand with the balloon **212**, thus taking on the shape of the balloon **212**. While the balloon **212** remains inflated, the heating source may be turned off, allowing the balloon **212**, stent and inflation fluid to cool. Once the stent has cooled, it may become stiff and retain its shape. The balloon **212** may then be deflated and removed, leaving the expanded stent in the correct drainage location. The stent may have a length sufficient to be disposed over both the proximal flared region **220a** and the distal flared region **220b** of the balloon **212**. When the stent is deployed, the flared regions **220a**, **220b** may create flared ends, or anti-migration features, in the stent. A first anti-migration feature may be positioned within the pseudocyst and a second anti-migration feature may be positioned within the gastrointestinal tract. It is contemplated that the size and shapes of the flared regions **220a**, **220b** and the intermediate region **224** may be selected to achieve a desired size and shape stent. For example, a length of the flared regions **220a**, **220b** along a longitudinal axis of the balloon **212** may be longer or shorter as desired. Other configurations are also contemplated.

(37) As discussed above, a stent may be formed or cross-linked from a shape memory polymer in its unexpanded shape or its expanded shape, as desired. When the stent is formed in the unexpanded shape, the expanded stent may take on the shape of the delivery balloon, as described above. FIG. 5 illustrates another illustrative shape memory polymer stent **300** that may be cross-

linked in the unexpanded shape or the expanded shape. The drainage stent **300** may include a flared proximal end **302**, a flared distal end **304** and a large diameter intermediate region **306** such that the inner diameter of the stent **300** varies over the length of the stent **300**. The proximal end **302** and the distal end **304** may be larger than the intermediate region **306** to prevent the drainage stent **300** from migrating once it has been deployed. When the stent **300** is cross-linked in an unexpanded, or generally tubular shape, a shaped deployment balloon, such as, but not limited to, balloons **20** or **212** may be utilized to achieve the desired expanded shape of the stent **300**. While not explicitly shown, a deployment balloon may have a barbell or curved shape to achieve the shape of stent **300**. It is further contemplated that stent **300** may be formed in the expanded configuration illustrated in FIG. 5. In such an instance, the stent **300** may be deployed using a “straight” balloon or one that does not include additional structural features (such as enlarged diameter portions) configured to change the shape of the expanded stent **300**.

(38) FIG. 6 illustrates another illustrative stent **350**. The stent **350** may be formed from a shape memory polymer. The stent **350** may be formed by injection molding SMP material. The stent **350** may be molded in its expanded configuration, with a large diameter and anti-migration features. The stent **350** may be cross-linked in this state, meaning that the expanded configuration is the remembered shape of the stent **350**. The stent **350** may include a proximal end **352** and a distal end **354**. The stent **350** may have a generally tubular intermediate region **356** extending between the proximal end **352** and the distal end **354**. The inner diameter of the stent **350** may be generally constant or uniform from the proximal end **352** to the distal end **354**. The stent **350** may further include a pair of retention features **358a**, **358b** positioned adjacent to the proximal and distal ends **352**, **354**. The retention features **358a**, **358b** may include inward facing barbs **360a**, **360b**. The barbs **360a**, **360b** may be sized and shaped to prevent the drainage stent **350** from migrating once it has been deployed between the pseudocyst and the gastrointestinal tract. The barbs **360a**, **360b** may be positioned on an outer surface of the stent **350**. The stent **350** may be formed having the final expanded shape and heated and deformed to be loaded onto a delivery balloon. It is contemplated that the delivery balloon may be generally straight. During deployment of the stent **350**, application of heat may cause the stent **350** to return to its remembered or expanded shape.

(39) FIG. 7 illustrates another illustrative self expanding stent **400**. The stent **400** may be formed from a shape memory polymer. The stent **400** may be molded in its expanded configuration, with a large diameter and anti-migration features. The stent **400** may be cross-linked in this state, meaning that the expanded configuration is the remembered shape of the stent **400**. The stent **400** may include a proximal end **402** and a distal end **404**. The stent **400** may have a generally tubular intermediate region **406** extending between the proximal end **402** and the distal end **404**. The stent **400** may further include a pair of retention features **408a**, **408b** positioned adjacent to the proximal and distal ends **402**, **404**. The inner diameter of the tubular intermediate portion **406** stent **400** may be generally constant or uniform from the proximal end **402** to the distal end **404** while an inner diameter of the retention features **408a**, **408b** may be larger than that of the intermediate portion **406**. However, it is contemplated that the inner diameter of the intermediate portion **406** may vary over the length thereof. The retention features **408a**, **408b** may be inwardly facing flares. For example, the proximal retention feature **408a** may increase in diameter from the proximal end **402** towards the distal end **404**. The distal retention feature **408b** may increase in diameter from the distal end **404** towards the proximal end **402**. This may allow the largest portion **410a**, **410b** of the retention features **408a**, **408b** to be positioned against the walls of the pseudocyst or gastrointestinal tract when the stent **400** is deployed. The retention features **408a**, **408b** may be sized and shaped to prevent the drainage stent **400** from migrating once it has been deployed between the pseudocyst and the gastrointestinal tract. The retention features **408a**, **408b** may be positioned on an outer surface of the stent **400**. The stent **400** may be formed having the final expanded shape and heated and deformed to be loaded onto a delivery balloon. It is contemplated that the delivery balloon may be generally straight. During deployment of the stent **400**, application

of heat may cause the stent **400** to return to its remembered or expanded shape.

(40) FIG. **8** illustrates another illustrative self expanding stent **450**. The stent **450** may be formed from a shape memory polymer. The stent **450** may be molded in its expanded configuration, with a large diameter and anti-migration features. The stent **450** may be cross-linked in this state, meaning that the expanded configuration is the remembered shape of the stent **450**. The stent **450** may include a proximal end **452** and a distal end **454**. The stent **450** may have a generally tubular intermediate region **456** extending between the proximal end **452** and the distal end **454**. The stent **450** may further include a pair of retention features **458a**, **458b** positioned adjacent to the proximal and distal ends **452**, **454**. The inner diameter of the stent **450** may be generally constant or uniform from the proximal end **452** to the distal end **454** while an inner diameter of retention features **458a**, **458b** may be larger than that of the stent **450**. However, it is contemplated that the inner diameter of the stent **450** may vary over the length thereof. The retention features **458a**, **458b** may be inwardly facing flares. In some instances, the retention features **458a**, **458b** may have a generally curved or cup-like profile. For example, the proximal retention feature **458a** may increase in diameter in a curved manner from the proximal end **452** towards the distal end **454**. The distal retention feature **458b** may increase in diameter in a curved manner from the distal end **454** towards the proximal end **452**. This may allow the largest portion **460a**, **460b** of the retention features **458a**, **458b** to be positioned against the walls of the pseudocyst or gastrointestinal tract when the stent **450** is deployed. The retention features **458a**, **458b** may be sized and shaped to prevent the drainage stent **450** from migrating once it has been deployed between the pseudocyst and the gastrointestinal tract. The retention features **458a**, **458b** may be positioned on an outer surface of the stent **450**. The stent **450** may be formed having the final expanded shape and heated and deformed to be loaded onto a delivery balloon. It is contemplated that the delivery balloon may be generally straight. During deployment of the stent **450**, application of heat may cause the stent **450** to return to its remembered or expanded shape.

(41) FIG. **9** illustrates another illustrative self expanding stent **500**. The stent **500** may be formed from a shape memory polymer. The stent **500** may be molded in its expanded configuration, with a large diameter and anti-migration features. The stent **500** may be cross-linked in this state, meaning that the expanded configuration is the remembered shape of the stent **500**. The stent **500** may include a proximal end **502** and a distal end **504**. The stent **500** may have a generally tubular intermediate region **506** extending between the proximal end **502** and the distal end **504**. The inner diameter of the stent **500** adjacent to the proximal and distal ends **502**, **504** may be larger than the inner diameter of the stent in the intermediate region **506**, although this is not required. The stent **500** may further include a pair of retention features **508a**, **508b** positioned adjacent to the proximal and distal ends **502**, **504**. The retention features **508a**, **508b** may be generally ring shaped protrusions extending from an outer surface of the stent **500**. In some instances, the retention features **508a**, **508b** may have an outer diameter that is larger than the outer diameter at any of the proximal end **502**, distal end **504**, or intermediate region **506**. The retention features **508a**, **508b** may be sized and shaped to prevent the drainage stent **500** from migrating once it has been deployed between the pseudocyst and the gastrointestinal tract. The stent **500** may be formed having the final expanded shape and heated and deformed to be loaded onto a delivery balloon. It is contemplated that the delivery balloon may be generally straight. During deployment of the stent **500**, application of heat may cause the stent **500** to return to its remembered or expanded shape.

(42) FIG. **10** illustrates another illustrative self expanding stent **550**. The stent **550** may be formed from a shape memory polymer. The stent **550** may be molded in its expanded configuration, with a large diameter and anti-migration features. The stent **550** may be cross-linked in this state, meaning that the expanded configuration is the remembered shape of the stent **550**. The stent **550** may include a proximal end **552** and a distal end **554**. The stent **550** may have a generally tubular intermediate region **556** positions between the proximal end **552** and the distal end **554**. The inner diameter of the stent **550** may be generally constant or uniform in the intermediate region **556**. The

stent **550** may further include a pair of retention features **558a**, **558b** positioned adjacent to the proximal and distal ends **552**, **554**. The retention features **558a**, **558b** may be outwardly facing flares. For example, the proximal retention feature **558a** may increase in diameter from the intermediate region **556** towards the proximal end **552**. The distal retention feature **558b** may increase in diameter from the intermediate region **556** towards the distal end **455**. In some instances, the taper **560a**, **560b** of the retention features **558a**, **558b** may be steep such that retention features **558a**, **558b** do not cause further dilation of the opening in the pseudocyst and/or gastrointestinal tract wall. The retention features **558a**, **558b** may be sized and shaped to prevent the drainage stent **550** from migrating once it has been deployed between the pseudocyst and the gastrointestinal tract. The stent **550** may be formed having the final expanded shape and heated and deformed to be loaded onto a delivery balloon. It is contemplated that the delivery balloon may be generally straight. During deployment of the stent **550**, application of heat may cause the stent **550** to return to its remembered or expanded shape. It is further contemplated that stent **550** may be formed in a generally tubular, unexpanded configuration. In this instance, stent **550** may be molded into the expanded configuration during deployment utilizing heat and a shaped delivery balloon having an outer surface profile corresponding to the desired final shape of the stent **550**.

(43) The materials that can be used for the various components of systems **10**, **200**, **300**, **350**, **400**, **450**, **500**, **550** (and/or other systems disclosed herein) may include those commonly associated with medical devices. For simplicity purposes, the following discussion makes reference to catheter shaft **12**. However, this is not intended to limit the systems and methods described herein, as the discussion may be applied to other components in systems **10**, **200**, **300**, **350**, **400**, **450**, **500**, **550**.

(44) Catheter shaft **12** and/or other components of systems **10**, **200**, **300**, **350**, **400**, **450**, **500**, **550** may be made from a metal, metal alloy, polymer (some examples of which are disclosed below), a metal-polymer composite, ceramics, combinations thereof, and the like, or other suitable material. Some examples of suitable metals and metal alloys include stainless steel, such as 304V, 304L, and 316LV stainless steel; mild steel; nickel-titanium alloy such as linear-elastic and/or super-elastic nitinol; other nickel alloys such as nickel-chromium-molybdenum alloys (e.g., UNS: N06625 such as INCONEL® 625, UNS: N06022 such as HASTELLOY® C-22®, UNS: N10276 such as HASTELLOY® C276®, other HASTELLOY® alloys, and the like), nickel-copper alloys (e.g., UNS: N04400 such as MONEL® 400, NICKELVAC® 400, NICORROS® 400, and the like), nickel-cobalt-chromium-molybdenum alloys (e.g., UNS: R30035 such as MP35-N® and the like), nickel-molybdenum alloys (e.g., UNS: N10665 such as HASTELLOY® ALLOY B2®), other nickel-chromium alloys, other nickel-molybdenum alloys, other nickel-cobalt alloys, other nickel-iron alloys, other nickel-copper alloys, other nickel-tungsten or tungsten alloys, and the like; cobalt-chromium alloys; cobalt-chromium-molybdenum alloys (e.g., UNS: R30003 such as ELGILOY®, PHYNOX®, and the like); platinum enriched stainless steel; titanium; combinations thereof; and the like; or any other suitable material.

(45) As alluded to herein, within the family of commercially available nickel-titanium or nitinol alloys, is a category designated “linear elastic” or “non-super-elastic” which, although may be similar in chemistry to conventional shape memory and super elastic varieties, may exhibit distinct and useful mechanical properties. Linear elastic and/or non-super-elastic nitinol may be distinguished from super elastic nitinol in that the linear elastic and/or non-super-elastic nitinol does not display a substantial “superelastic plateau” or “flag region” in its stress/strain curve like super elastic nitinol does. Instead, in the linear elastic and/or non-super-elastic nitinol, as recoverable strain increases, the stress continues to increase in a substantially linear, or a somewhat, but not necessarily entirely linear relationship until plastic deformation begins or at least in a relationship that is more linear than the super elastic plateau and/or flag region that may be seen with super elastic nitinol. Thus, for the purposes of this disclosure linear elastic and/or non-super-elastic nitinol may also be termed “substantially” linear elastic and/or non-super-elastic nitinol.

(46) In some cases, linear elastic and/or non-super-elastic nitinol may also be distinguishable from

super elastic nitinol in that linear elastic and/or non-super-elastic nitinol may accept up to about 2-5% strain while remaining substantially elastic (e.g., before plastically deforming) whereas super elastic nitinol may accept up to about 8% strain before plastically deforming. Both of these materials can be distinguished from other linear elastic materials such as stainless steel (that can also can be distinguished based on its composition), which may accept only about 0.2 to 0.44 percent strain before plastically deforming.

(47) In some embodiments, the linear elastic and/or non-super-elastic nickel-titanium alloy is an alloy that does not show any martensite/austenite phase changes that are detectable by differential scanning calorimetry (DSC) and dynamic metal thermal analysis (DMTA) analysis over a large temperature range. For example, in some embodiments, there may be no martensite/austenite phase changes detectable by DSC and DMTA analysis in the range of about -60 degrees Celsius (C) to about 120° C. in the linear elastic and/or non-super-elastic nickel-titanium alloy. The mechanical bending properties of such material may therefore be generally inert to the effect of temperature over this very broad range of temperature. In some embodiments, the mechanical bending properties of the linear elastic and/or non-super-elastic nickel-titanium alloy at ambient or room temperature are substantially the same as the mechanical properties at body temperature, for example, in that they do not display a super-elastic plateau and/or flag region. In other words, across a broad temperature range, the linear elastic and/or non-super-elastic nickel-titanium alloy maintains its linear elastic and/or non-super-elastic characteristics and/or properties.

(48) In some embodiments, the linear elastic and/or non-super-elastic nickel-titanium alloy may be in the range of about 50 to about 60 weight percent nickel, with the remainder being essentially titanium. In some embodiments, the composition is in the range of about 54 to about 57 weight percent nickel. In other embodiments, a superelastic alloy, for example a superelastic nitinol can be used to achieve desired properties.

(49) In at least some embodiments, portions or all of system **10, 200, 300, 350, 400, 450, 500, 550** may also be doped with, made of, or otherwise include a radiopaque material. Radiopaque materials are understood to be materials capable of producing a relatively bright image on a fluoroscopy screen or another imaging technique during a medical procedure. This relatively bright image aids the user of systems **10, 200, 300, 350, 400, 450, 500, 550** in determining its location. Some examples of radiopaque materials can include, but are not limited to, gold, platinum, palladium, tantalum, tungsten alloy, polymer material loaded with a radiopaque filler, and the like. Additionally, other radiopaque marker bands and/or coils may also be incorporated into the design of system **10, 200, 300, 350, 400, 450, 500, 550** to achieve the same result.

(50) In some embodiments, a degree of Magnetic Resonance Imaging (MRI) compatibility is imparted into systems **10, 200, 300, 350, 400, 450, 500, 550**. For example, catheter shaft **12** or portions thereof, may be made of a material that does not substantially distort the image and create substantial artifacts (i.e., gaps in the image). Certain ferromagnetic materials, for example, may not be suitable because they may create artifacts in an MRI image. Catheter shaft **12** or portions thereof, may also be made from a material that the MRI machine can image. Some materials that exhibit these characteristics include, for example, tungsten, cobalt-chromium-molybdenum alloys (e.g., UNS: R30003 such as ELGILOY®, PHYNOX®, and the like), nickel-cobalt-chromium-molybdenum alloys (e.g., UNS: R30035 such as MP35-N® and the like), nitinol, and the like, and others.

(51) Some examples of suitable polymers that may be suitable for use in systems **10, 200, 300, 350, 400, 450, 500, 550** may include polytetrafluoroethylene (PTFE), ethylene tetrafluoroethylene (ETFE), fluorinated ethylene propylene (FEP), polyoxymethylene (POM, for example, DELRIN® available from DuPont), polyether block ester, polyurethane (for example, Polyurethane 85A), polypropylene (PP), polyvinylchloride (PVC), polyether-ester (for example, ARNITEL® available from DSM Engineering Plastics), ether or ester based copolymers (for example, butylene/poly(alkylene ether) phthalate and/or other polyester elastomers such as HYTREL®

available from DuPont), polyamide (for example, DURETHAN® available from Bayer or CRISTAMID® available from Elf Atochem), elastomeric polyamides, block polyamide/ethers, polyether block amide (PEBA, for example available under the trade name PEBAX®), ethylene vinyl acetate copolymers (EVA), silicones, polyethylene (PE), Marlex high-density polyethylene, Marlex low-density polyethylene, linear low density polyethylene (for example REXELL®), polyester, polybutylene terephthalate (PBT), polyethylene terephthalate (PET), polytrimethylene terephthalate, polyethylene naphthalate (PEN), polyetheretherketone (PEEK), polyimide (PI), polyetherimide (PEI), polyphenylene sulfide (PPS), polyphenylene oxide (PPO), poly paraphenylene terephthalamide (for example, KEVLAR®), polysulfone, nylon, nylon-12 (such as GRILAMID® available from EMS American Grilon), perfluoro (propyl vinyl ether) (PFA), ethylene vinyl alcohol, polyolefin, polystyrene, epoxy, polyvinylidene chloride (PVdC), poly(styrene-b-isobutylene-b-styrene) (for example, SIBS and/or SIBS 50A), polycarbonates, ionomers, biocompatible polymers, other suitable materials, or mixtures, combinations, copolymers thereof, polymer/metal composites, and the like.

(52) Those skilled in the art will recognize that the present disclosure may be manifested in a variety of forms other than the specific embodiments described and contemplated herein. Accordingly, departure in form and detail may be made without departing from the scope of the present disclosure as described in the appended claims.

Claims

1. A stent delivery catheter, comprising: a catheter shaft having a proximal end and a distal end; wherein the catheter shaft further comprises an inner member and an outer member; an inflatable balloon having an inner region and an outer surface, the inflatable balloon disposed adjacent to the distal end of the catheter shaft; a cutting electrode disposed on the distal end of the inner member of the catheter shaft; and a self-expanding stent disposed about the catheter shaft and adjacent to the distal end, the stent having a non-expanded state and an expanded state, wherein the stent in the expanded state comprises a proximal retention feature and a distal retention feature, wherein the distal retention feature increases in diameter from a distal end of the self-expanding stent towards a proximal end of the self-expanding stent, wherein the proximal retention feature increases in diameter from the proximal end of the self-expanding stent towards the distal end of the self-expanding stent, and wherein at least one of the proximal or distal retention features are configured to engage an interior portion of a pseudocyst or a gastric structure to prevent migration of the self-expanding stent.
2. The stent delivery catheter of claim 1, wherein the expanded state of the stent includes a flared proximal end region and a flared distal end region.
3. The stent delivery catheter of claim 2, wherein the flared proximal end region and the flared distal end region each comprise an inwardly facing flare portion.
4. The stent delivery catheter of claim 1, wherein an outer diameter of the inflatable balloon varies along the inflatable balloon.
5. The stent delivery catheter of claim 4, wherein the inflatable balloon has a first region having a first outer diameter, a second region having a second outer diameter, and a third region having a third outer diameter, the second diameter smaller than the first and third diameters, and wherein the stent in the expanded state comprises a shape substantially corresponding to the first region, the second region, and the third region of the inflatable balloon.
6. The stent delivery catheter of claim 5, wherein the stent is disposed in the non-expanded state about the outer surface of the inflatable balloon along the first region, the second region, and the third region.
7. The stent delivery catheter of claim 1, wherein the stent is configured to create a drainage path between a gastrointestinal tract lumen and a pseudocyst.

8. A method for delivering a stent between a first body lumen and a second body lumen, the method comprising: advancing a stent delivery catheter to a target location within the first body lumen, the catheter comprising: a catheter shaft extending from a proximal end to a distal end; wherein the catheter shaft further comprises an inner member and an outer member; an inflatable balloon disposed along the catheter shaft proximate the distal end of the catheter shaft; a cutting electrode positioned at the distal end of the inner member of the catheter shaft; and a stent disposed about the catheter shaft proximate the distal end of the catheter shaft, the stent having a first unexpanded configuration and a second expanded configuration; energizing the cutting electrode in contact with the first body lumen to create an opening from the first body lumen; energizing the cutting electrode in contact with the second body lumen to create an opening to the second body lumen; inflating the inflatable balloon within the opening from the first body lumen and the opening to the second body lumen; and deploying the stent to the second expanded configuration within the opening from the first body lumen and the opening to the second body lumen, wherein the stent in the expanded configuration comprises a proximal retention feature and a distal retention feature, wherein the distal retention feature increases in diameter from a distal end of the self-expanding stent towards a proximal end of the self-expanding stent, wherein the proximal retention feature increases in diameter from the proximal end of the self-expanding stent towards the distal end of the self-expanding stent, and wherein at least one of the proximal or distal retention features are configured to engage an interior portion of a pseudocyst or a gastric structure to prevent migration of the self-expanding stent.

9. The method of claim 8, wherein in the second expanded configuration, the stent comprises a proximal retention member, a distal retention member, or both.

10. The method of claim 9, wherein the proximal retention member, the distal retention member, or each of the proximal retention member and the distal retention member comprises a flared end region.

11. The method of claim 9, wherein deploying the stent results in the proximal retention member being positioned within the first body lumen, the distal retention member being positioned within the second body lumen, or both.

12. The method of claim 8, wherein an outer diameter of the inflatable balloon varies along the inflatable balloon.

13. The method of claim 12, wherein the stent in the second expanded configuration substantially corresponds to the varying outer diameter of the inflatable balloon.

14. The method of claim 8, wherein the stent comprises a bioabsorbable shape memory polymer configured to expand when exposed to body temperature.

15. A method for delivering a stent between a first body lumen and a second body lumen, the method comprising: advancing a stent delivery catheter to a target location within the first body lumen, the catheter comprising: a catheter shaft having a proximal end and a distal end; wherein the catheter shaft further comprises an inner member and an outer member; an inflatable balloon disposed along the catheter shaft adjacent to the distal end of the catheter shaft; a cutting electrode disposed on the distal end of the inner member of the catheter shaft; and a stent disposed about the catheter shaft, the stent having a non-expanded state and an expanded state; energizing the cutting electrode in contact with the first body lumen to create an opening from the first body lumen; energizing the cutting electrode in contact with the second body lumen, through the opening from the first body lumen, to create an opening to access the second body lumen; inflating the inflatable balloon within one or both of the opening from the first body lumen and the opening to the second body lumen; and transforming the stent to the expanded state across the opening from the first body lumen and through the opening to the second body lumen, wherein the stent in the expanded state comprises a proximal retention feature and a distal retention feature, wherein the distal retention feature increases in diameter from a distal end of the self-expanding stent towards a proximal end of the self-expanding stent, wherein the proximal retention feature increases in diameter from the

proximal end of the self-expanding stent towards the distal end of the self-expanding stent, and wherein at least one of the proximal or distal retention features are configured to engage an interior portion of a pseudocyst or a gastric structure to prevent migration of the self-expanding stent.

16. The method of claim 15, wherein in the expanded state, the stent comprises a proximal retention member, a distal retention member, or both.

17. The method of claim 16, wherein the proximal retention member, the distal retention member, or each of the proximal retention member and the distal retention member comprises a flared end region.

18. The method of claim 16, wherein transforming the stent to the expanded state across the openings includes positioning the proximal retention member within the first body lumen, positioning the distal retention member within the second body lumen, or both.

19. The method of claim 15, wherein an outer diameter of the inflatable balloon varies along the proximal end to the distal end of the inflatable balloon.

20. The method of claim 15, wherein the first body lumen is a gastrointestinal tract lumen and the second body lumen is a pseudocyst.
